

Joins

Agenda and reminders

- HW 2 due Friday on **GitHub classroom**
 - Commit early and often, let me know if you have any technical problems
 - Make sure to submit both the `.qmd` **and** `.md` files
- Today: combining data from different tables

Data stored in multiple tables

The `nycflights13` package contains information on flights from NYC airports in 2013. The data is stored across several data frames:

- `airlines`: information on each airline
- `airports`: information on each airport
- `flights`: information on each flight
- `planes`: information on each plane
- `weather`: hourly weather data

Question: What is the advantage of storing this data in multiple tables, instead of one BIG table?

Data stored in multiple tables

- Databases often contain different tables to store different information
- For example, a healthcare database could contain the following tables:
 - patients
 - doctors
 - offices
 - insurance

Joining tables

1 patients

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang

1 doctors

	provider	location
1	Dr. Foyle	Greensboro
2	Dr. Renard	Winston-Salem
3	Dr. Zhang	Winston-Salem

I want to add location information to the `patient` table.
What should the resulting table look like?

Left join

```
1 patients
```

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang

```
1 doctors
```

	provider	location
1	Dr. Foyle	Greensboro
2	Dr. Renard	Winston-Salem
3	Dr. Zhang	Winston-Salem

```
1 patients |>  
2   left_join(doctors, join_by(provider))
```

	age	insurance	provider	location
1	23	Aetna	Dr. Zhang	Winston-Salem
2	47	BCBS	Dr. Foyle	Greensboro
3	38	Medicaid	Dr. Zhang	Winston-Salem

Left join

```
1 patients |>  
2   left_join(doctors, join_by(provider))
```

	age	insurance	provider	location
1	23	Aetna	Dr. Zhang	Winston-Salem
2	47	BCBS	Dr. Foyle	Greensboro
3	38	Medicaid	Dr. Zhang	Winston-Salem

- Left joins are useful for adding additional information to a table
- Left joins (generally) keep the same rows as the initial dataframe (`patients`), and add more columns
- `join_by` specifies how to link the tables

Joining tables

Flights information:

```
# A tibble: 3 × 5
```

	time_hour <dtm>	origin <chr>	dest <chr>	tailnum <chr>	carrier <chr>
1	2013-01-01 05:00:00	EWK	IAH	N14228	UA
2	2013-01-01 05:00:00	LGA	IAH	N24211	UA
3	2013-01-01 05:00:00	JFK	MIA	N619AA	AA

Weather information

```
# A tibble: 3 × 4
```

	origin <chr>	time_hour <dtm>	temp <dbl>	wind_speed <dbl>
1	EWK	2013-01-01 01:00:00	39.0	10.4
2	EWK	2013-01-01 02:00:00	39.0	8.06
3	EWK	2013-01-01 03:00:00	39.0	11.5

Question: What if I want to get information about the weather for each flight?

Left joins

```
1 flights |>
2   left_join(weather, join_by(origin, time_hour))
```

A tibble: 6 × 7

	time_hour <dtm>	origin <chr>	dest <chr>	tailnum <chr>	carrier <chr>	temp <dbl>	wind_speed <dbl>
1	2013-01-01 05:00:00	EWR	IAH	N14228	UA	39.0	12.7
2	2013-01-01 05:00:00	LGA	IAH	N24211	UA	39.9	15.0
3	2013-01-01 05:00:00	JFK	MIA	N619AA	AA	39.0	15.0
4	2013-01-01 05:00:00	JFK	BQN	N804JB	B6	39.0	15.0
5	2013-01-01 06:00:00	LGA	ATL	N668DN	DL	39.9	16.1
6	2013-01-01 05:00:00	EWR	ORD	N39463	UA	39.0	12.7

Joining with different names

Suppose our tables looked like this:

1 patients				1 doctors		
	age	insurance	provider		name	location
1	23	Aetna	Dr. Zhang	1	Dr. Foyle	Greensboro
2	47	BCBS	Dr. Foyle	2	Dr. Renard	Winston-Salem
3	38	Medicaid	Dr. Zhang	3	Dr. Zhang	Winston-Salem

How would we specify the columns to link the tables?

Joining with different names

Suppose our tables looked like this:

```
1 patients
```

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang

```
1 doctors
```

	name	location
1	Dr. Foyle	Greensboro
2	Dr. Renard	Winston-Salem
3	Dr. Zhang	Winston-Salem

```
1 patients |>  
2   left_join(doctors, join_by(provider == name))
```

	age	insurance	provider	location
1	23	Aetna	Dr. Zhang	Winston-Salem
2	47	BCBS	Dr. Foyle	Greensboro
3	38	Medicaid	Dr. Zhang	Winston-Salem

Another join

Patients in the system:

1 patients

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang
4	54	Humana	Dr. Renard

Accepted insurance:

1 insurance

	company	phone
1	Anthem	800-676-2583
2	BCBS	877-258-3334
3	Kaiser	800-810-4766
4	Medicaid	877-201-3750

Suppose I want insurance information only for the patients who have an accepted insurance. What should the final table look like?

Inner join

Patients in the system:

```
1 patients
```

	age	insurance	provider
1	23	Aetna	Dr. Zhang
2	47	BCBS	Dr. Foyle
3	38	Medicaid	Dr. Zhang
4	54	Humana	Dr. Renard

Accepted insurance:

```
1 insurance
```

	company	phone
1	Anthem	800-676-2583
2	BCBS	877-258-3334
3	Kaiser	800-810-4766
4	Medicaid	877-201-3750

```
1 patients |>
```

```
2   inner_join(insurance, join_by(insurance == company))
```

	age	insurance	provider	phone
1	47	BCBS	Dr. Foyle	877-258-3334
2	38	Medicaid	Dr. Zhang	877-201-3750

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)

For next time, read:

- Chapter 26.2 in *R for Data Science*